

Upload CSV

Choose a CSV File

patient\_records.csv  
158.0B

+

Rule Engine Specification

The agent dynamically scans any dataset:

1. Universal Checks:
- o Duplicate rows.

o Null or missing values in any column.

o Mixed data types.
2. Text Formatting:
- o Outer whitespace warnings.
3. Numeric Outliers:
- o Mathematical outlier detection using the Interquartile Range (IQR).

o Negative checks on positive count fields.
4. Standard Constraints:
- o If `Age` is found: bounds checked [0, 100].

o If `email` is found: regex format validated.

o Missing checks in `name` and `roll number`.

API Secrets Configuration

Configure your OpenRouter API Key by creating:

- A `.streamlit/secrets.toml` file with `OPENROUTER_API_KEY="..."`

OR

A `.env` file in the root folder with `OPENROUTER_API_KEY="..."`



AI Data Quality Agent

Upload any CSV dataset to scan for data quality issues and anomalies. Our automated rule engine dynamically checks for missing values, duplicates, mixed types, email formats, and outliers, while **OpenRouter (Nex-AGI)** diagnoses patterns and writes custom Python cleanup scripts!



Dataset Preview

|   | patient_id | full_name   | age | blood_type |
|---|------------|-------------|-----|------------|
| 0 | P001       | John Doe    | 45  | O+         |
| 1 | P002       | Jane Smith  | -12 | A-         |
| 2 | P003       | Bob Johnson | 29  | XYZ        |
| 3 | P004       | Alice Brown | 34  | B+         |
|   |            |             |     |            |



Detected Column Attributes

| Column Name    | Inferred Type     | Completeness (%) | Unique Values |
|----------------|-------------------|------------------|---------------|
| patient_id     | Categorical       | 100.0%           |               |
| full_name      | Categorical       | 100.0%           |               |
| age            | Numeric (Integer) | 100.0%           |               |
| blood_type     | Categorical       | 100.0%           |               |
| heart_rate_bpm | Numeric (Integer) | 100.0%           |               |

## Detected Errors

- ⚠ Detected 2 data quality warning(s)/anomaly(ies) in dataset:
- Row 2: Age is -12 (must be between 0 and 100)
  - Column 'heart\_rate\_bpm': Statistical outlier(s) detected: Row 4 (value: 250)



## AI Suggestions



```
df.loc[~age_valid_mask, AGE_COL] =
age_median
else:
    raise ValueError('No valid age
values found to use for imputation.')

# 2) Fix statistical heart-rate
outliers using the IQR rule.
# The observed bad value is 250 bpm,
so it is replaced with the median non-
outlier heart rate.
hr_non_null = df[HR_COL].dropna()
q1 = hr_non_null.quantile(0.25)
q3 = hr_non_null.quantile(0.75)
iqr = q3 - q1
lower_bound = q1 - 1.5 * iqr
upper_bound = q3 + 1.5 * iqr
```